Identification of an Unknown – “B”

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Introduction

The goal of this lab is to identify unknown sample B using NMR, GCMS, and FTIR spectroscopy. Understanding and interpreting these spectroscopy readings are necessary skills because these are efficient methods of identification used in many different fields. Usually, these spectroscopy readings are a faster means of determining what a compound is compared to the more time-consuming melting point or boiling point methods used in labs throughout the semester.

Body

Broadly, for each of these three methods, energy is passed through the molecules of the unknown sample to monitor how they react. Depending on how much energy is absorbed, or how much energy changed after the spin of the nucleus, the resulting readings can illustrate behaviors that are intrinsic and unique for every compound and element.

1. **NMR**

   According to Appendix II, "In this technique, a broad pulse of electromagnetic radiation excites all the H or C nuclei simultaneously, resulting in a continuously decreasing oscillation caused by the decay of excited nuclei back to their stable energy distribution" (85). The frequencies that are collected are put through a computer program that uses the Fourier transform to convert the free-induced decay (FID) into a readable spectrum. Depending on the number of frequencies that are read, the compound will be more complex.

2. FTIR

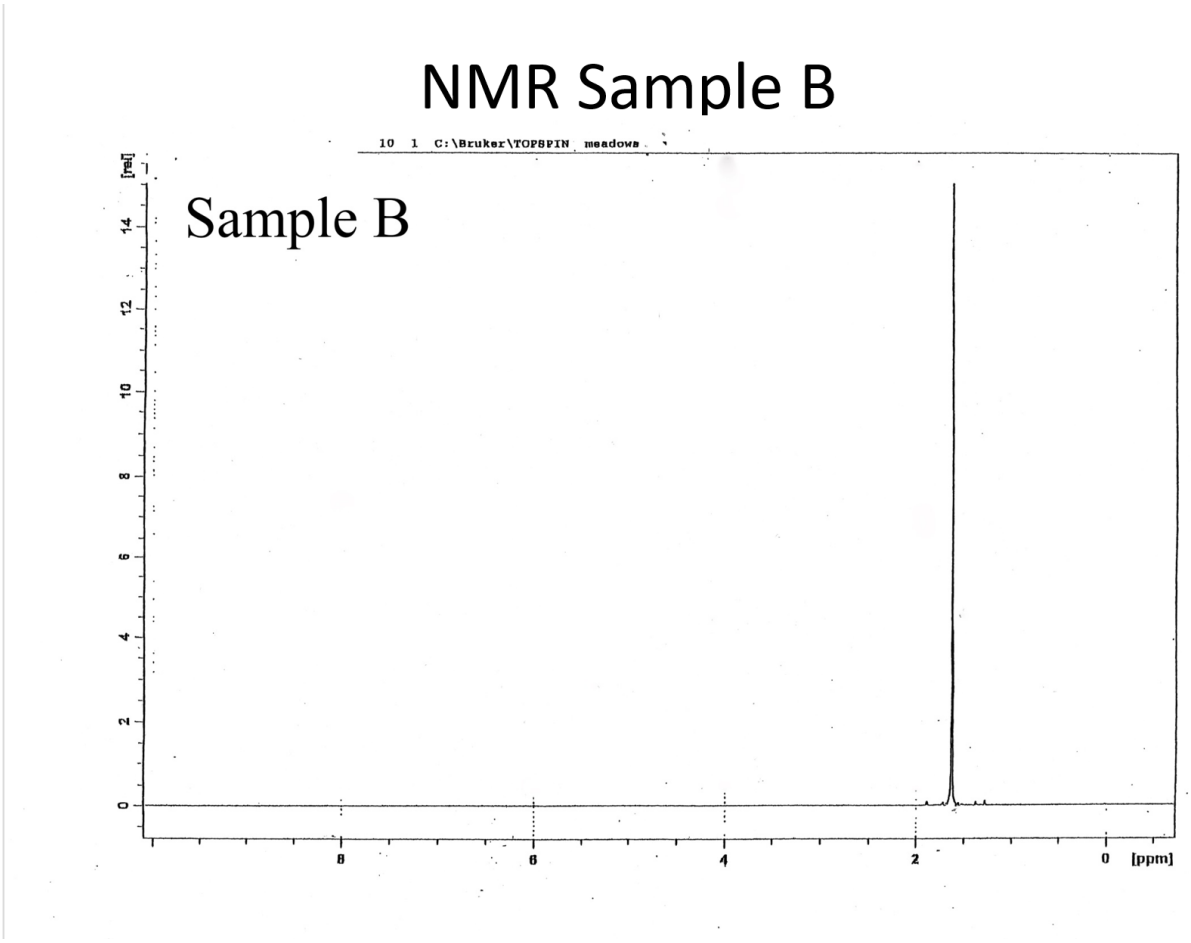
   Infrared (IR) spectroscopy is a method in which energy is passed through organic compounds to observe how much is absorbed at different positions. FTIR is a great method for determining the "presence or absence of simple functional groups". The intensity of

absorption illustrates molecular vibrations, which can indicate stretching vibration, symmetric stretching vibration, and bending vibration, as well as bond order and atomic mass relations. Additionally, overtone bands can show when “fundamental vibrations produce intense absorption peaks” (137). There are three key regions to inspect when analyzing IR spectra, which are the functional group region, the fingerprint region, and the aromatic region.

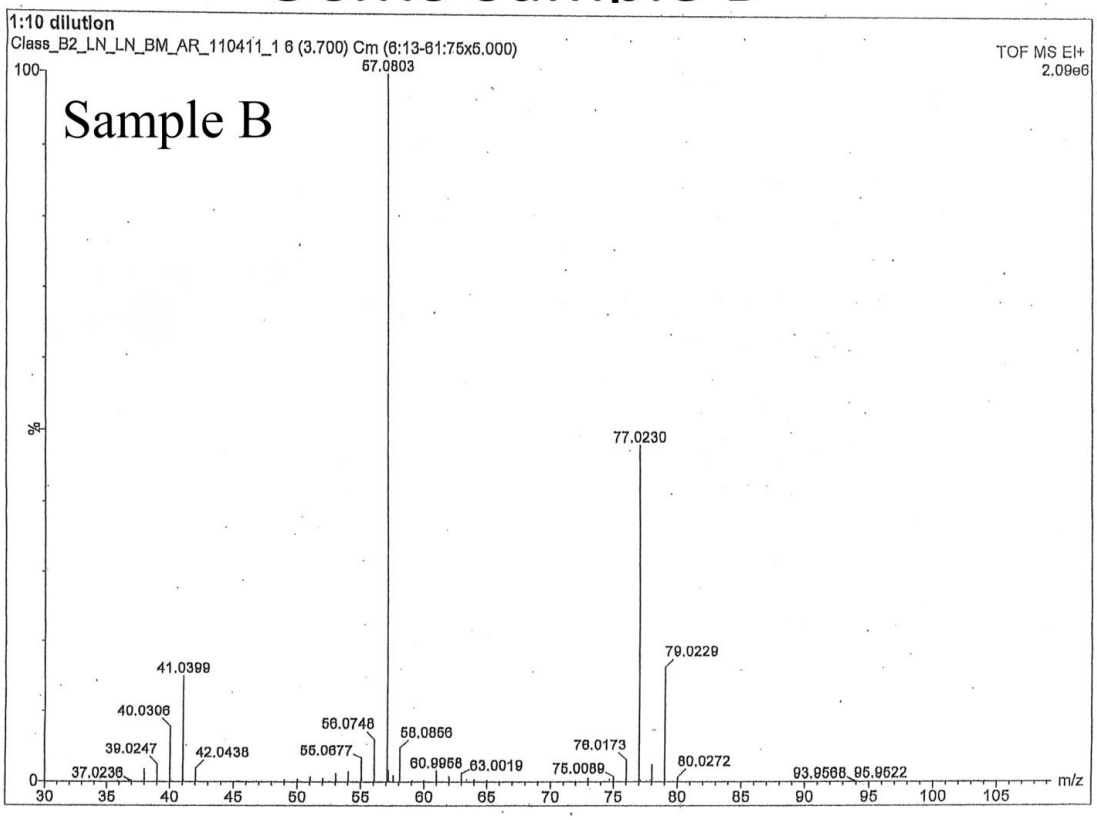
3. MS

For mass spectrometers, the compound is “converted into a gas-phase ion”, then the ions are sorted by a mass analyzer, and “the sorted ions generate an electric current at the detector that corresponds to the number of ions … of each *m/z* value, creating a mass spectrum”(157). A graph of this mass spectrum is a good resource for determining isotope abundance, the presence of fragment ions, and the base molecular formula of the compound using the Rule of Thirteen.

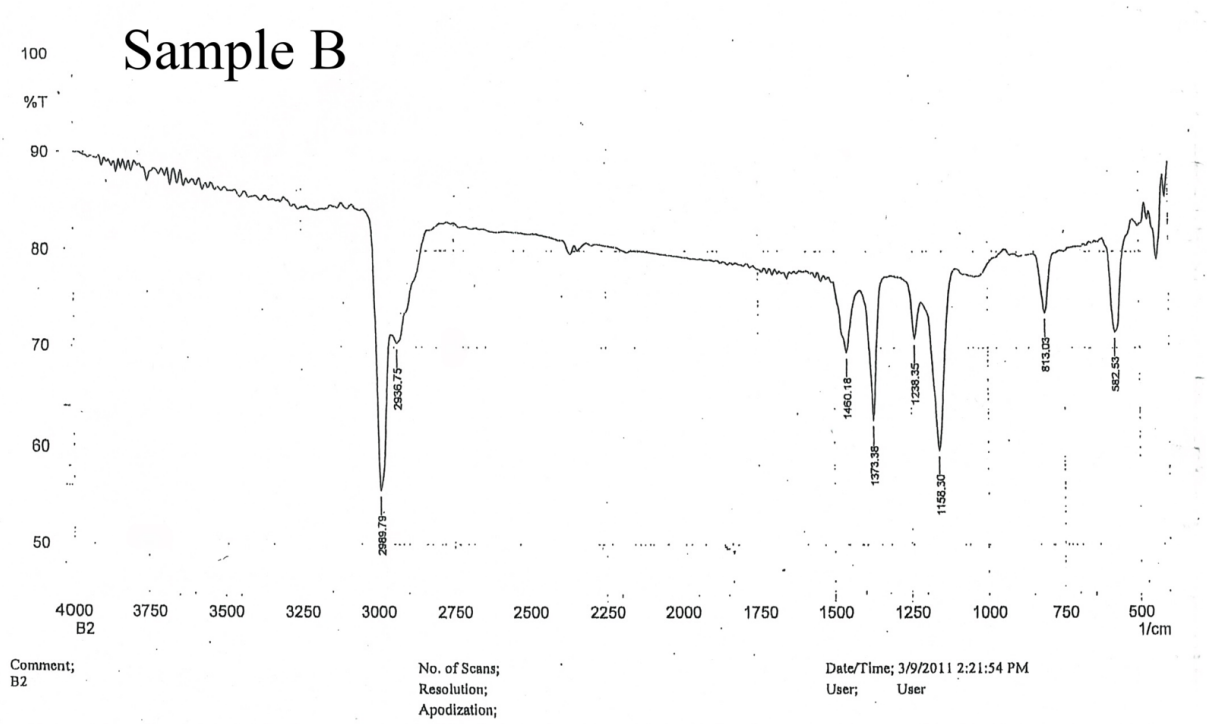
Data


NMR Sample B
10 1 C:\Bruker\TOPSPIN meadows
Sample B
[rel]
14
12
10
8
6
4
2
0
8
6
4
2
0
[ppm]

GCMS Sample B



FTIR Sample B



Analysis

For the NMR reading, a single peak is observed at just below 2ppm. According to Figure II.8 in the appendix, that could fall in the range of alkanes. It also falls in edges of the ranges for alkynes, allylic compounds, alcohols, and amines, but since there are no other peaks that could be in those ranges, the compound being an alkane is more likely. The Table II-1 confirms this assumption, since it lists that the chemical shift of alkanes is from 0.8-1.9.

For the GCMS reading, the base peak was observed at 57 m/z, and the molecular ion peak was observed at 79 m/z. The Rule of Thirteen can be used, in which the m/z of the molecular ion peak is divided by 13 to find the *n* value in the formula C*n*H*m,* and the remainder

of this division is added to the *n* value to get the *m* value. Dividing 79 m/z by 13 returns a value of 6.07, or an n value of 6 and an m value of 1. This results in a molecular formula of C6H7.

For the FTIR reading, a large peak was observed in the 3000-2850cm^-1 region, as well as a few peaks the fingerprint region and the aromatic region. The large peak in the functional group region falls on the range of C-H and O-H bonds, according to the Figure III-7 (140). Additionally, the peaks in the fingerprint region are at 1373 and 1155 cm^-1, which fall within the range of a bond order of 1, like for C-C, C-O, and C-N bonds. The most significant characteristics of this reading are the intensity at 582 cm^-1, which indicates a C-Cl stretch, and that the largest peak is at position 2969 cm^-1, indicating strong intensity of C-H stretch and there is sp^3 hybridization.

Conclusion

With the assumption that the Rule of Thirteen from the GCMS reading is a leading characteristic, the molecular formula contains C6H7, and the FTIR indicates the presence of chlorine. A possible identity with these assumptions is 1-Chlorocyclohexa-1,3-diene, which meets these requirements. However, this molecule does not have the characteristic sp^3 hybridization, and it has double bonds when the FTIR reading suggested a higher likelihood of a presence of single bonds.

If the Rule of Thirteen calculation was an incorrect assumption, another possible identity for the unknown could be tert-Butylchloride (1-chloro-3-methylpropane). Although it doesn't have as many carbons as were calculated, it does have sp^3 hybridization and only single bonding.

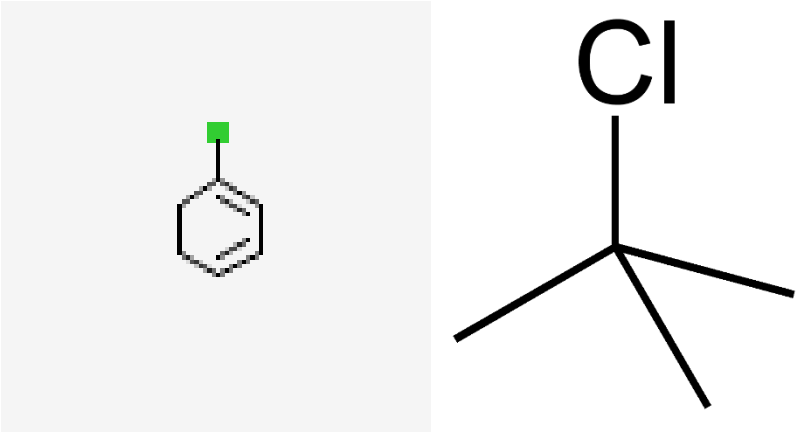


Figure 1. The image on the left is 1-Chlorocyclohexa-1,3-diene, where the green box indicates the location of the chlorine. The image on the right illustrates 1-chloro-3-methylpropane.

References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.